

William Baldwin Smith V

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Location: Charlottesville, VA

EDUCATION

- **University of Virginia** Charlottesville, VA
Doctor of Philosophy in Electrical Engineering Aug 2026 - present
- **University of Virginia** Charlottesville, VA
Master of Engineering in Mechanical Engineering Aug 2021 - May 2023
Courses: Experimental Robotics, Autonomous Mobile Robots, Robotic Autonomy, Robotic Manipulators, Linear Control Systems, Dynamical Systems, Analytical Dynamics, Optimization Methods, Machine Learning, Advanced Machine Learning.
- **University of Virginia** Charlottesville, VA
Bachelor of Science in Mechanical Engineering Aug 2018 - May 2021

SKILLS SUMMARY

- **Languages:** C/C++, Python, MATLAB, Java, R
- **Libraries:** OpenCV, GStreamer, PyTorch, NumPy, Matplotlib
- **Tools:** ROS/ROS2, Linux, Git, Docker, Simulink, SOLIDWORKS, Autodesk Inventor, Ansys, UV
- **Hardware:** Arduino, Raspberry Pi, NVIDIA Jetson, ESP32

EXPERIENCE

- **Applied Research Associates, Inc.** Raleigh, NC
Staff Scientist I, ISR, Analytical Spatial Intelligence Research Team Oct 2023 - Jul 2026
 - **Real Time Target Re-identification for Surveillance with Unmanned Aerial Vehicles:**
 - * Led team of 4 engineers to convert research-grade Python code to deployable C++ code for edge devices by introducing and enforcing coding standards with complete style sheets and targeted unit testing.
 - * Developed sensor interfaces to retrieve state data from the flight controller and format as MISB 0601 metadata packets to be pushed to an RTSP camera server.
 - * Deployed and tested a full system stack on various custom and commercial UAV platforms with successful re-identification, reducing duplicate targets by over 7x and accurately geolocating targets within 5 meters.
 - **GPS-Denied Localization for Unmanned Aerial Vehicles:**
 - * Implemented a visual inertial odometry system to predict the 6 DoF displacement of a UAV.
 - * Integrated online UAV video and metadata streaming with programmatic planning and control capabilities.
 - * Developed GStreamer pipelines to mux video and MISB 0601 formatted metadata into an MPEG-TS file or set up an RTSP server.
 - **Urban Geolocalization of Wild Photos:**
 - * Improved map creation with parallel computing, to increase generation speed by over 100x, and efficient compression techniques, to reduce global map size by 75 percent.
 - * Led integration effort to convert research code into a deployable system with a multi-threaded optimization in Python, leading to over 10x speed boost.
 - * Researched a system to estimate camera intrinsics and extrinsics of images by ensembling multiple learning-based estimators.
 - **Passive Ranging of Long Range Targets:**
 - * Investigated machine learning-based depth estimation models to obtain passive ranging for every pixel in the image.
 - * Implemented a PCA-based correction to restore depth scale through consecutive panned or zoomed images.
 - **Localization in a GPS-Denied Urban Area:**
 - * Optimized and cross-compiled algorithms for an embedded processor to directly interface with onboard sensors.
 - * Developed an asynchronous processing pipeline to pass data from sensors to different stages of the algorithm.
 - * Implemented a particle filter for state estimation to constrain the search region, leading to improved accuracy and speed.
 - * Conducted live demonstrations for customers with an ATAK integration to show the GPS and estimated position.
- **University of Virginia** Charlottesville, VA
Research Assistant Mar 2021 - Mar 2023
 - **High Precision Mapping:**
 - * Developed a method to improve robotic mapping of buildings with millimeter resolution and high accuracy.
 - * Researched methods to quantify mapping quality to improve inspections by minimizing environmental uncertainty.

- * Improved path planning techniques to guarantee reliable autonomous mapping with online trajectory optimization to ensure desired resolution and accuracy.
- * Demonstrated high pose accuracy compared to COLMAP by generating NeRF models with improved smoothness and scene realism.

- **Heterogeneous Robot Cooperation:**

- * Formulated a system for heterogeneous multi-robot cooperation in a human-centric environment for performing inspection and maintenance in a shipboard environment.
- * Assigned different tasks to robots to minimize mission time by sharing beliefs to increase collaboration and understanding of the environment and situation.

- **Specialized Robot Development:**

- * Designed and built unmanned ground vehicles with varying embodiments to carry a robotic manipulator.
- * Implemented low-level and high-level control algorithms to allow basic autonomy with various tasks.

- **NASA Langley Research Center**

Hampton, VA

Engineering Intern

June 2017 - July 2017

- **Custom Rover Design:**

- * Designed and developed a radio-controlled rover for the Autonomy Incubator as part of a team of three interns.
- * Constructed to meet target specifications, including: carry at least 50 pounds, weigh under 15 pounds, and be stable.

ACADEMIC PROJECTS

- **Mobile Manipulation Motion Planning:** Trajectory planning for mobile manipulators to result in smooth and continuous motions. Researched a whole-body planner for mobile manipulators using Lie theory and optimization. Modeled robots using screw joints and Lie theory to find kinematic constraints. Formulated the optimization with pose constraints and other smoothing costs to guarantee smooth and continuous motion. Presented at the Mobile Manipulation: Emerging Opportunities and Contemporary Challenges workshop at Robotics Science and Systems (RSS) 2025. (2023-2025)
- **Multi-Stage High Precision Mapping:** Developed a method to improve robotic mapping of buildings with millimeter resolution and high accuracy. Researched methods to quantify mapping quality to improve inspections by minimizing environmental uncertainty. Improved path planning techniques to guarantee reliable autonomous mapping with online trajectory optimization to ensure desired resolution and accuracy. Presented at IEEE IEEE International Symposium on Safety, Security, and Rescue Robotics (SSRR) 2022 and published in Robotics 2023. (2021-2023)
- **Additive Manufacturing with Extruded Clay via a Mobile Manipulator:** Created an autonomous system for additive manufacturing using mobile manipulators. The clay medium required precise coordination of the mobile base and manipulator to ensure the structure was built appropriately without knocking over the structure. Collaborated with other students and researchers in other departments. (2022)
- **Autonomous Campus Vehicles:** Designed an autonomous golf cart by modifying a standard electric golf cart with drive-by-wire controls. Implementation of SLAM for a sparse, outdoor environment by fusing depth cameras and LiDAR to create a globally accurate map for navigation. Integration of convoying between the leading vehicle and following vehicles with shared location beliefs and visual correction. (2021-2022)
- **Visual Servoing for Shipboard Inspection and Maintenance:** Developed systems to perform basic manipulation tasks with a human in the loop supervisor for maintenance. The main goal was to identify and localize emergency buttons in a naval environment. The robot would then travel to the location and identify the available buttons for a remote user to select before executing the motion to achieve the task. (2021)

HONORS AND AWARDS

- Mobile Manipulation: Emerging Opportunities and Contemporary Challenges Workshop Presenter at Robotics Science and Systems 2025 (June 2025)
- University of Virginia Mechanical and Aerospace Engineering Student Seminar Speaker (March 2023)
- Session Chair of Mapping in Unstructured Environment at 2022 IEEE International Symposium on Safety, Security, and Rescue Robotics (November 2022)
- University of Virginia Link Lab Flash Talk Finalist (October 2022)
- University of Virginia Mechanical Engineering Research Fair Presenter (October 2022)
- University of Virginia Engineering Research Symposium Podium Speaker (3rd Place) (April 2022)

PUBLICATIONS

- [1] **Smith, W.**, Singh, S., Rudy, J., Guan, Y., “Whole Body Planning of Mobile Manipulators Leveraging Lie Theory based Optimization,” en, in *RSS 2025 Workshop: Mobile Manipulation: Emerging Opportunities & Contemporary Challenges*, Jun. 2025. [Online]. Available: <https://openreview.net/forum?id=7ZRclFFHSK¬eId=7ZRclFFHSK>.
- [2] **Smith, W.**, Qin, Y., Singh, S., Burke, H., Furukawa, T., Dissanayake, G., “A multistage framework for autonomous robotic mapping with targeted metrics,” *Robotics*, vol. 12, no. 2, 2023, issn: 2218-6581. DOI: 10.3390/robotics12020039. [Online]. Available: <https://www.mdpi.com/2218-6581/12/2/39>.
- [3] **Smith, W.**, Qin, Y., Furukawa, T., Dissanayake, G., “Autonomous Robotic Map Refinement for Targeted Resolution and Local Accuracy,” in *2022 IEEE International Symposium on Safety, Security, and Rescue Robotics (SSRR)*, 2022, pp. 130–137. DOI: 10.1109/SSRR56537.2022.10018686.